Esther Rituerto-González

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WORK EXPERIENCE

Sept 2021 – Present

VISITING RESEARCH SCHOLAR – CHAIR OF EMBEDDED INTELLIGENCE FOR HEALTHCARE AND WELLBEING (EIHW), AUGSBURG UNIVERSITÄT, under the supervision of **Prof. Dr.-Ing. habil. Björn Schuller**.

+ **sustAGE Project** (EU Horizon 2020 Research & Innovation Action) Smart Environments for Person-centered Sustainable Work and Well-being

- Sept 2021 - Dec 2021: **German Academic Exchange Service (DAAD) Short Term Grant Scholarship**

- Dec 2021- Feb 2022: Supplementary **Short-stay Mobility Grants** for beneficiaries of the **University Teacher in Training program (FPU**)

Oct 2020 – Present

UNIVERSITY TEACHER IN TRAINING, RESEARCH SCHOLARSHIP (FPU 19/00448) – MINISTRY OF SCIENCE, INNOVATION AND UNIVERSITIES (MICIU), SPAIN

+ **EMPATIA-CM Project** (Ref: Y2018/TCS-5046) by **UC3M4Safety**, to develop '*BINDI*', a wearable solution for detecting, preventing and combating violence against Women through speech and audio analysis and machine learning algorithms
+ **Gender Studies Institute (IEG),** Carlos III University of Madrid (UC3M)
+ **Multimedia Processing Group (GPM),** Signal Theory and Communications Department (TSC), UC3M

Febr 2020 – Oct 2020

PREDOCTORAL RESEARCH PERSONNEL IN TRAINING by COMUNIDAD DE MADRID YEI PREDOCTORAL PROGRAM (PEJD-2019-PRE/TIC-16295) – UC3M

+ **EMPATIA-CM** Project**, UC3M4Safety** & **IEG** & **GPM** (TSC)

April 2019 – Jan 2020

RESEARCH SUPPORT TECHNICIAN CONTRACTED BY EMPATIA-CM (REF: Y2018/TCS-5046) – UC3M

+ **EMPATIA-CM** Project**, UC3M4Safety** & **GPM** (TSC)

October 2018 – Dic 2018

RESEARCH SUPPORT TECHNICIAN CONTRACTED BY SAPIENS (REF: TEC2017-84395-P) – UC3M

+ **SAPIENS Project** (Saliency and Attention: rePresentation, Interpretation and EmergeNce) & **GPM** (TSC)

November 2017 – Oct 2018

MASTER'S SCHOLARSCHIP – UC3M

+ **GPM** (TSC)
+ **Teaching Assistant at Signal Theory and Communications Department**, **GPM**
    - *Multimedia,* BSc in Computer Science Engineering
    - *Digital Image Processing,* BSc in Audiovisual Systems Engineering
    - *Acoustics,* BSc in Audiovisual Systems Engineering

January 2017 – June 2017 **INTERNSHIP STUDENT** – IMDEA Networks Madrid, Spain

+ Providing support to Engineering, Operations and Communications Departments

EDUCATION

Nov 2019 – Present **PhD IN MULTIMEDIA AND COMMUNICATIONS** – UC3M

+ Pursuing a doctoral thesis titled "Multimodal affective computing in wearable devices with applications in the detection of gender-based violence"

Sept 2017 – June 2018 **MEng: MASTER IN MULTIMEDIA AND COMMUNICATIONS** (in English) – UC3M

+ Specialization in signal and data processing, speech analysis, and machine learning

Sept 2013 – May 2017 **DEGREE IN AUDIOVISUAL SYSTEMS ENGINEERING** – UC3M

+ Multimedia services development in communication networks, image, audio and video signal processing, acoustic conditioning of enclosures

COURSES

July 2021 **DEEPLEARN 2021 SUMMER SCHOOL** (40H) – IRDTA. *Las Palmas de Gran Canaria, Spain.*

June 2021 **TICKET TO OPEN SCIENCE & OPEN SCIENCE CAFÉS** (24H) – UC3M. *Held online.*

June 2021 **PROGRESS WORKSHOP (PROmotinG DiveRsity in Signal ProcESSing)** (16H) - ICASSP 2021 where awarded an IEEE Signal Processing Society Professional Development Grant

January - June 2021 **COURSE TRAINING IN GENDER EQUALITY, GENDER PERSPECTIVE AND GENDER VIOLENCE** (75H) - Rosa San Segundo, Professor of the Institute of Gender Studies (IEG), *UC3M*

July 2020 **LXMLS: LISBON MACHINE LEARNING SCHOOL** (50H) – Instituto Superior Técnico (IST), Lisbon, Portugal. *Held virtually.*

July 2020 **ADVANCED TRAINING FOR ONLINE TEACHING** (10H) - UC3M

June 2020 **EMOTIONS AND STRESS MANAGEMENT** (8H) – Raquel Reyes (Instituto de Psicología de Emoción y Salud), UC3M

Oct 2019 – May 2020 **DRAMA COURSE** (60H) – School of The Arts, UC3M

July 2019 **SUMMER SCHOOL ON SPEECH TECHNOLOGIES** – by Red Temática de Tecnologías del Habla (RTTH), Vicomtech San Sebastián, Spain

November 2019 **INCLUSIVE LANGUAGE FOR EQUAL COMMUNICATION** (5H) – Mercedes Bengoechea, UC3M

June – Nov 2019 **EMOTIONAL INTELLIGENCE, ASSERTIVENESS AND PERSONALITY** (28H) –UC3M

Dec 2018 – Feb 2019 **CERTIFICATE IN MACHINE LEARNING** (55H) – Stanford University, Coursera

Sept 2015 – May 2018 **DUBBING COURSE** (250H) – "*Escuela de Doblaje de Madrid*", Spain

+ Acting and vocal skills formation

Sept 2013 – May 2014 **COURSE ON THE DEVELOPMENT OF TRANSFERABLE SKILLS** (20H) – UC3M

+ Time management, team work, leadership and creativity

Julio 2012 **ENGLISH LANGUAGE COURSE ABROARD (C2 LEVEL)** (72H) – EDUCATION FIRST (EF) London Waterloo EF School, London, UK

PUBLICATIONS

May 2021, **Article** **End-to-end Recurrent Denoising Autoencoder Embeddings for Speaker Identification.** Rituerto-González, Esther; Peláez-Moreno, Carmen. *Neural Computing and Applications (JCR **Q1**), SPRINGER*. DOI:*10.1007/s00521-021-06083-7*

December 2020, **Proceedings** **Reinterpretation of Emotional Labelling and Acoustic Events Augmentation of Data for the Automatic Detection of Gender-Based Violence.** Rituerto-González, Esther; Luis-Mingueza, Clara; Peláez-Moreno, Carmen. *IBERSPEECH 2020 Proceedings*. DOI:*10.21437/IberSPEECH.2021-13*

July 2020, **Proceedings** **A Hybrid Data Fusion Architecture for BINDI: a Wearable Solution to Combat Gender-based Violence.** Rituerto-González, Esther; Miranda, Jose A.; Canabal, Manuel F.; Lanza-Gutierrez, José M.; Peláez-Moreno, Carmen and López-Ongil, Celia. *10th International Conference on Multimedia Communications, Services & Security (MCSS 2020).* DOI:*10.1007/978-3-030-59000-0_17*

March 2019, **Article** **Data Augmentation for Speaker Identification under Stress Conditions to Combat Gender-based Violence.** Rituerto-González, Esther; Mínguez-Sánchez, Alba; Peláez-Moreno, Carmen and Gallardo-Antolín, Ascensión. *Applied Sciences Journal: Special Issue "IberSPEECH 2018 Speech and Language Technologies for Iberian Languages" (JCR **Q2**) MDPI.* DOI:*10.3390/app9112298*

November 2018, **Proceedings** **Speaker Recognition under Stress Conditions.** Rituerto-González, Esther; Gallardo-Antolín, Ascensión and Peláez-Moreno, Carmen. *IBERSPEECH 2018 Proceedings* p15-19. DOI:*10.21437/IberSPEECH.2018-4*

CONFERENCES

June 2021 **VI International Congress of Young Researchers with a Gender Perpective** at UC3M, Getafe, where presented **The Influence of Sex and Gender in Artificial Intelligence**

April 2021 **I Internacional Congress "R+D+I Technology for gender equality: Solutions, perspectives and challenges".** at UC3M, Getafe, where presented **Fear Detection in Speech**

March 2021	**IBERSPEECH 2020** Conference in Valladolid, Spain by ISCA, held online, where presented **Reinterpretation of Emotional Labelling and Acoustic Events Augmentation of Data for the Automatic Detection of Gender-Based Violence**
October 2020	**10th International Conference on Multimedia Communications, Services & Security (MCSS 2020)** Held on-line, where it was presented **A Hybrid Data Fusion Architecture for BINDI: a Wearable Solution to Combat Gender-based Violence**
June 2020	**V International Congress of Young Researchers with a Gender Perpective** at UC3M, Getafe, where it was presented **Recognizing the victim's voice in gender-based violence situations: A machine learning approach**
June 2019	**IV International Congress of Young Researchers with a Gender Perpective** at UC3M, Getafe, where presented **Wearable Device to Combat Gender-based Violence**
November 2018	**IBERSPEECH 2018** Conference in Barcelona, Spain by ISCA, with an ISCA Travel Grant where presented **Speaker Recognition under Stress Conditions**

HONORS AND AWARDS

March 2019	**IV Edition Pilar Azcárate Award UC3M** in Gender Violence Research: Master's Thesis Modality with **Speaker Recognition under Stress Conditions**
September 2017	**Honors Thesis UC3M** in **Machine Learning for Visual Attention Models in the Gastronomic Field,** Bachelor Thesis
September 2013	**UC3M Excellence Scholarship** for the best records in High School

OTHER ACHIEVEMENTS

June 2021	**Bachelor Thesis Tutor** for **Shallow Learning Applied to the Classification of Emotional States from Biosignals** in *Carlos III University of Madrid* by **Lois Riobó Mayo**
June 2021	**Virtual Conference Assistance** for **ICASSP 2021** (46th International Conference on Acoustics, Speech, and Signal Processing). Toronto, Canada. *Held virtually*
January 2021	**Virtual Workshop Assistance** for **IEEE SLT 2020** (IEEE Spoken Langage Technology Workshop: deep learning and beyond). Shenzhen, China. *Held virtually*
November 2020	**Virtual Workshop Assistance** for **ODYSSEY 2020** (The Speaker and Language Recognition Workshop). Tokyo, Japan. *Held virtually*
June 2020	**Virtual Conference Assistance** for **ICASSP 2020** (45th International Conference on Acoustics, Speech, and Signal Processing). Barcelona, Spain. *Held virtually*
March 2019	**Career Break** for *one month* to take time off to decide on a the career path that really fit my personal goals and improve my English level in **Edinburgh, United Kingdom (UK)**

LANGUAGE SKILLS

Spanish	**NATIVE**

German	**BASIC LEVEL**	
	2022	German as Foreign Language, Augsburg Universität (**A2** Level)
	2021	Basic Level Title by Official School of Languages Leganés (**A1** Level)
English	**ADVANCED LEVEL *(WRITTEN AND SPOKEN)***	
	2013	CERTIFICATE IN ADVANCED ENGLISH (**C1**, Grade B)
	2012	**C2** Level in EF Cambridge English Level Test
	2012	Advanced Level of English by Official School of Languages Talavera (**C1**)
	2011	FIRST CERTIFICATE IN ENGLISH (**B2**, Grade A)
French	**BASIC LEVEL *(WRITTEN AND SPOKEN)***	

IT SKILLS

Programming languages	**PYTHON** (PYTORCH, KERAS, TENSORFLOW, SK-LEARN)	●●●●○
	MATLAB	●●●●○
	LATEX	●●●●○
	JAVA and C	●●●○○
	HTML, CSS, JAVASCRIPT	●●●○○
OS	**WINDOWS, LINUX, macOS**	●●●●○
Office suite	**MICROSOFT OFFICE 365, OPEN OFFICE**	●●●●○
Photo and video editing	**ADOBE PHOTOSHOP**	●●●○○
	SONY VEGAS	●●●●○